

AGUIO, JOHN WAYNE CARL P.

OOP

BSIT 2-3

ACTIVITY # 1 - VARIABLES

PART A - VARIABLE IDENTIFICATION TABLE

Variable Name	Data Type	Variable Type	stored in
owner	String	non-primitive	Heap
balance	double	primitive	Heap
acc1	Account	non-primitive	Stack
acc2	Account	non-primitive	Stack
acc3	Account	non-primitive	Stack

PART B - EXPLANATION

Statement	What Happens?	Memory Impact
Account acc1 = new Account("Carlos", 5000);	Creates a new Account object with owner Carlos and balance 5000	Object created in heap, acc1 stores reference in stack
Account acc2 = new Account("Diana", 7000);	Creates another Account object with owner Diana and balance 7000	Second object created in heap
Account acc3 = acc1;	acc3 now refers to the same object as acc1	both acc1 and acc3 point to the same heap object
acc2.balance = 8000;	updates Diana's balance to 8000	value inside acc2 object changes in heap
acc3.owner = "Miguel";	change owner of the object referenced by acc1 and acc3	The stored object's owner changes to Miguel

PART C - OUTPUT

Miguel 5000.0

Diana 8000.0

Miguel 5000.0